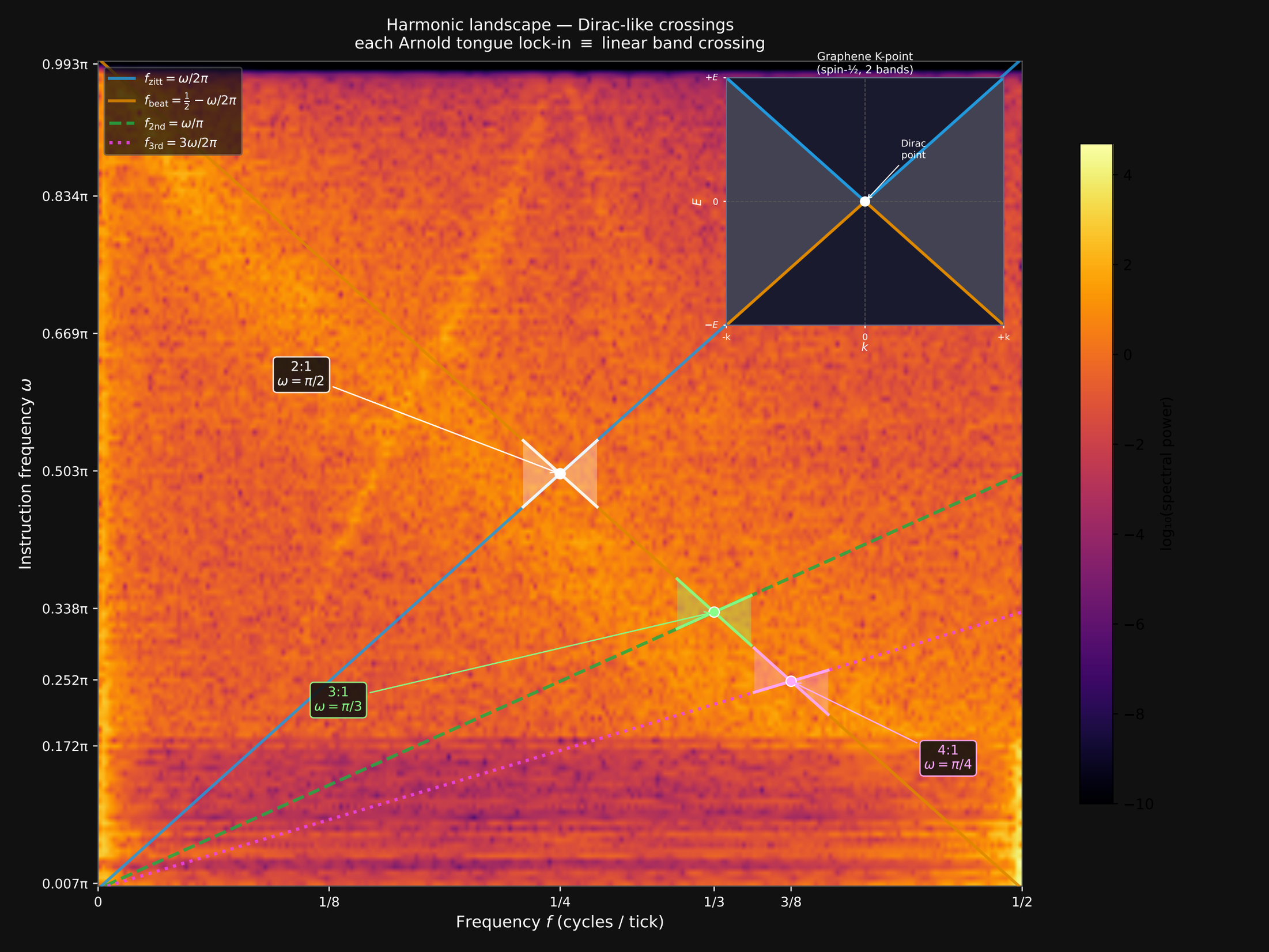
Harmonic landscape — Dirac-like crossings
each Arnold tongue lock-in ≡ linear band crossing
$f_{\text{zitt}} = \omega/2\pi$
$f_{\text{beat}} = \frac{1}{2} - \omega/2\pi$
$f_{\text{2nd}} = \omega/\pi$
$f_{\text{3rd}} = 3\omega/2\pi$
2:1
$\omega = \pi/2$
3:1
$\omega = \pi/3$
4:1
$\omega = \pi/4$
Graphene K-point
(spin-½, 2 bands)
Dirac point
+E
0
−E
E
-k
0
+k
k
0.993π
0.834π
0.669π
0.503π
0.338π
0.252π
0.172π
0.007π
Instruction frequency ω
0
1/8
1/4
1/3
3/8
1/2
Frequency f (cycles / tick)
4
2
0
−2
−4
−6
−8
−10
log₁₀(spectral power)